Moving the Mean, Not the Best: What Inference-Time Interventions and Weight Consolidation Buy in Open-Ended Generation

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Abstract

A companion study [7] found that interrupting a base language model with a periodic new subject raises judged surprise but composes into nothing: no integrated documents, and more verified candidates without a better best. Here we ask what would make local gains compose, giving the loop, one piece at a time, what a discovering mind has: schematic memory, a standing question, selection, a verifier inside the stream, and consolidation of its training-verifier-selected candidates into its weights. Every battery was pre-registered. At inference time one thing worked: a model-written schematic recap re-injected at each interruption raises document-level integration (+1.65 against verbatim context, $p = 0.001$; most of the headline is the reset it rides on, +0.45, $p = 0.13$, against reset alone), and nothing moves development above about 2 on a 0–10 scale. A verifier whose verdicts are written into the stream is imitated rather than used: 16.4 fabricated verdict lines per notebook. Consolidation is different in kind: in one five-cycle lineage, LoRA on value-filtered candidates shifts held-out generation toward value (-1.7 points, $p = 0.008$, in a sequential-analysis sense; -3.1 against a random-consolidation control, $p = 0.004$), while the best observed candidate never improves and the per-candidate rate of better-than-classic candidates on far families falls from 10% to at most 3.9% (absolute yields, resting on one to five events, run the other way). Inference-time stimulation reorganizes a fixed prior; weight consolidation moves its mass toward the known good, at a per-candidate cost in the tails that the registered independent lineages now running must confirm.

1 Introduction

Ono Filho [7] dismantled a cognitively inspired generation loop over 24 conditions and found that most of its effect on judged novelty lives in one operation, the *interruption*: a new subject injected every few hundred tokens into a stream whose literal repetition is damped. On windows of fresh generated text the interruption raises judged surprise by 1.2 to 1.4 points over habituation alone, on three base models, a post-trained one and two genres, and a pre-registered replication confirms it. The same study closed with three negatives. Read as whole documents, none of the streams builds anything (no arm above 2.5 on integration, development, coherence or surprise, 0–10, judged on 4,500 tokens with the injected text removed); a judge-gated review that decides when not to interrupt does not change that; and on online bin packing, where a deterministic verifier replaces the judge, the interruption multiplies the valid, distinct candidate heuristics a base model writes three- to fourfold without raising the quality of the best. The interruption is a variation operator, and variation without selection is organized noise.

This paper takes up the question that study left: what would make local gains compose? We proceed by giving the loop, one piece at a time, the things a mind that discovers has and the loop did not, and measuring each under the protocol that the companion study showed to be necessary (generated-only windows, fresh-only estimates, a document-level reading, the premise as the unit of inference, a deterministic verifier where one exists). The pieces are (i) a *schematic* memory, in which the model carries a compressed recap of its own stream instead of the verbatim text that entrains it [1]; (ii) a *standing question*, in which the interruption points into the stream's own unresolved matter rather than away from it [16]; (iii) *selection*, a FunSearch-style evolution over the candidates the interruption multiplies [9]; (iv) a *verifier inside the stream*, whose verdict on each candidate is written back into the notebook; and (v) *consolidation*, STaR/ReST cycles in which the model is fine-tuned (LoRA) on its own training-verifier-selected candidates and generates again [3, 11, 17], followed by preference-based repulsion [8] and by quality-diversity selection [4, 5] designed to keep the tails. Every battery was pre-registered before it ran; the pre-registrations, their dated amendments and the results are in the laboratory notebook released with the code.

Contributions. (1) A schematic recap, written by the model and re-injected at each interruption, is the first intervention of the program to raise the document-level reading: integration +1.65 over the verbatim context and the highest integration score of any arm, at no cost on fresh windows. (2) A standing question raises integration and development modestly; its combination with the recap does not raise development further. Across all injections development stays near 2/10: the *progression wall*. (3) Selection over the interruption's candidates and a verifier inside the stream both leave the ceiling where it was; the latter produces a new phenomenon, fabricated verifier verdicts, which we quantify. (4) Consolidation by LoRA on training-verifier-selected candidates moves the candidate distribution on never-seen variants of the trained family toward value ($p = 0.008$ after five cycles; vs a random-consolidation control $p = 0.004$), keeps diversity over five cycles, leaves the ceiling where it was and does not transfer to other families; anchored repulsion from the worst moves the mass onto the best classic heuristic exactly; two tail-preserving objectives restore functional diversity and not the tails, and across every consolidation the rate of better-than-classic candidates on new families is highest in the untrained base. (5) A reading of the whole: the bit distance between a fixed prior and a target is not shortened by inference-time stimulation; weight consolidation moves the mass, not the frontier, and pays for the mass with the tails.

2 Related work

Self-training on one's own outputs. STaR [17] fine-tunes a model on its own rationales that led to correct answers; ReST [3] and ReST-EM [11] alternate sampling, filtering by a reward and supervised fine-tuning, and report gains that saturate after a few iterations. Our attract arm is that recipe at office scale (an 8B model, forty examples per cycle, a rank-8 adapter) with three additions: a same-size random-consolidation control, which separates "trained on its own outputs" from "trained on what the verifier approved"; transfer measured on variants and families that never entered any training set; and the rate of candidates better than the classic heuristics, which is where the cost of self-training shows.

Our consolidation result is close to the finding that reward training concentrates probability on existing trajectories, raising low-k performance while large-k coverage stays or falls [15]. The literature also contains the counterexamples that sharpen its scope: boundary-guided curriculum RL reports raising both pass@1 and pass@256 above the base [2], and representation-based exploration bonuses improve pass@k in post-training [13]; both add what is absent here, external guidance or directed exploration beyond the model's own rollouts.

Preference optimization. DPO [8] optimizes a policy against a frozen reference from chosen/rejected pairs. Our repulsion arms use it with the model's own worst candidates, or the clones of the known optimum, as rejected examples; unanchored it degenerates, anchored with a supervised term on the chosen side it sharpens the policy onto the known optimum. Both behaviours are known failure and success modes of preference optimization; what we add is their effect on the tails of a candidate distribution measured by a verifier.

Quality-diversity. MAP-Elites [5] and novelty search [4, 12] keep the best solution per behavioural niche rather than the best overall. Our qd arm consolidates one elite per niche of the verifier's per-instance behaviour profile; it keeps functional diversity and, in this regime, loses value and the tails alike.

Verified search. FunSearch [9] and AlphaEvolve [6] pair an untrained proposal model with a hard evaluator and many samples, without fine-tuning the proposer. Our results are consistent with that choice: the untrained prior has the highest rate of better-than-classic candidates, and every consolidation we ran lowered it.

Memory, tension, and the judge. Schematic memory as reconstruction [1]; unfinished tasks as retained tension [16]; compression progress as intrinsic reward [10]. On the instrument, the companion study documents what a windowed LLM judge cannot see (injected text, self-copy beyond its horizon, windows against wholes); the fabricated verdicts of Section 5 are the same entrainment acting on a verifier's output placed in the prompt.

3 Setup

Generators and loop. Unless stated, the generator is Qwen3-30B-A3B-Base (mixture of experts, 8-bit, MLX on an Apple M5 Pro); battery C uses Qwen3-8B-Base (8-bit) because it is fine-tuned. A cell is one premise (the ten premises of the companion study) continued for 4,500 tokens with the end-of-text token masked; habituation (windowed repetition penalty, window 512, factor 1.15) is on in every arm; interruptions fire on a clock every 300 tokens. Arms differ in what is injected and in whether the context is kept or rebuilt.

Judging. Windows of generated text only (96 tokens, starting 32 after any injected text, never crossing an injection), judged by Claude Opus 5 ($k = 5$, median) on surprise, connection and coherence with 600 tokens of context; windows flagged as self-copy (half their 12-token shingles seen earlier in the stream) are excluded from the fresh-only estimates. The whole stream, injected text removed, is judged once per cell ($k = 3$) on integration, development, coherence and surprise. The premise is the unit: cell means, bootstrap intervals over cells, exact sign-flip permutation tests on paired cells, one-sided where pre-registered.

The problem with a verifier. Online bin packing in the FunSearch form: the model writes a notebook of `priority(item, remaining)` functions; the verifier packs instances and reports the mean excess over the lower bound. Batteries S and V use the ten distribution variants of the companion study (item ranges within $[0.1, 0.8]$, where best fit is near-optimal); battery C moves to a family of small-item distributions in which best fit is measurably beatable (Section 6).

arm (Qwen3-30B-A3B-Base, period 300)	integration	development	coherence	surprise
habituation, no interruption	1.40	1.10	1.50	1.40
verbatim context (clock300)	0.80	0.40	1.50	0.50
reset + new subject	2.00	1.60	2.20	2.50
schema: reset + recap + new subject	**2.70**	1.80	2.30	2.70
question: preserved + open question	1.50	1.10	1.90	1.00
agenda: reset + recap + open question	2.50	**2.00**	1.90	2.60

Table 1: Document-level reading (whole 4,500-token stream, injected text removed; Opus 5, $k = 3$; cell means over ten premises, 0–10). The schematic recap gives the highest integration of the program; no arm passes 2.0 on development.

4 Memory and tension: what the loop remembers and what it is asked

Arms. *Verbatim* (`clock300`): the context is kept across interruptions. *Reset* (`reset300`): the cache is rebuilt from the premise and the injected sentence only. *Schema* (`schema300`): at each interruption the model is asked, by a completion cue on its last 1,200 tokens, to write a two- to three-sentence past-tense recap; the cache is rebuilt from the premise, the recap ("*By then, this much had happened: ...*") and the new subject. The recap counts as injected text (never inside a judged window; removed from the document). *Question* (`anomaly300`): the context is kept, and the injection is the open question the model itself extracts from its last 1,200 tokens ("*But one question remained: ...*"). *Agenda* (`agenda300`): reset context, recap and question in the same injection. An anti-copy ladder retries the model's recap or question at higher temperatures when it reproduces the previous one (it tends to), and falls back to the plain rotation when it cannot; the oracle acts only after 200 tokens of stream.

Results. Table 1 gives the document-level reading. Pre-registered contrasts (one-sided, exact paired permutation over ten premises): M2, schema > verbatim on (integration+development)/2: $+1.65$ $[+1.15, +2.25]$, $p = 0.001$, positive on every dimension (integration $+1.90$, development $+1.40$, coherence $+0.80$, surprise $+2.20$; 10/10 premises on integration and surprise). M3, question > neutral subject: $+0.70$ $[+0.40, +1.05]$, $p = 0.004$, with integration and development each $+0.70$ ($p = 0.03$). M1, schema > reset on the composite: $+0.45$ $[-0.15, +1.05]$, $p = 0.13$, not supported; decomposed, integration $+0.70$ $[+0.10, +1.30]$ and development $+0.20$: the recap gives the model something to integrate and nothing to advance. M4, on fresh windows the recap costs nothing (surprise $+0.15$, connection -0.08, coherence -0.27, all n.s.; self-copy 0%; the schema arm's fresh windows, 3.90 / 3.22 / 6.60, are the best local scores of the program). The agenda arm (A1, primary: agenda > schema on development) gives $+0.20$ $[-0.40, +0.80]$, $p = 0.38$; the question changes the cast of each restart, not its progression (document judge: "restarts the premise over and over with new casts and genres").

Reading. Remembering the schema beats re-reading the text: the verbatim context, which entrains the model on thousands of tokens of its own past, reads worst as a whole; a reset reads better; a reset carrying a compressed memory reads best. But three injections (schema, question, agenda), all pre-registered, move integration and never development above about 2/10. We call this the progression wall.

5 Selection and the verifier

Battery S: the interruption inside a selection loop. A FunSearch-style evolution (10 variants $\times$ 8 generations $\times$ 16 samples per generation; the prompt shows the current population; paired by variant) with and without the interruption's angle line injected into the prompt. S1 (gain of the champion on held-out instances): +0.0013 [-0.0015, +0.0047], $p = 0.38$. S2 (cumulative distinct valid candidates): 89.5 vs 95.1, $p = 0.86$ against the hypothesis. S3 (generation of first escape from best fit): +0.5, n.s. Under selection the interruption adds nothing, not even the diversity it added in a single stream: the population prompt already supplies what the interruption supplied.

Battery V: the verifier inside the stream. The notebook of the companion study's battery B, generated in chunks; whenever a `priority` function closes, the harness scores it on the training instances and writes the verdict into the notebook as a comment ("`# Verifier: priority_x scored mean excess 0.0812 on the training instances (worse than best fit, 0.0752); no improvement.`"). A second arm adds, every 300 tokens, a deterministic scoreboard ("`# So far: N functions tried; the best is ...; best fit sits at ...`") and a value-anchored open question. Ten variants $\times$ two seeds; baselines are battery B's plain and angle arms re-scored in text order. V1 (held-out gain, feedback vs angle): +0.0003, $p = 0.48$. V2 (candidates improving on the notebook's own training best, the verifier's measure of progression): 0.1 vs 0.3 per notebook, opposite direction. V4 (scoreboard adds to feedback): nothing; the scoreboard arm writes fewer functions than every other arm (1.1 per notebook). Progression is near zero in every arm (0.1–0.3 improvements per notebook), with the standing caveat that best fit is near-optimal on these distributions.

Fabricated verdicts. The model wrote on average 16.4 "`# Verifier:`" lines per notebook in the feedback arm (6.2 with the scoreboard) against about three real ones: verdicts with invented scores and "new best of this notebook" that the harness never issued. Injecting the *format* of a reward signal into the context makes a base model continue a highly predictable textual pattern present in its context; whether the fabrication functions as reward or as mere format imitation is untested (no fabricated verdict was produced by the harness; their correlation with true scores and their effect on subsequent candidates are open analyses). The rule is narrower than "keep verdicts out of the prompt": do not place forgeable verifier syntax in the same generative channel without provenance (a non-generable channel, a nonce, reserved tokens) and without measuring imitation.

6 Consolidation: moving the prior

Design (pre-registered 2026-08-21, extensions 2026-08-22). Qwen3-8B-Base (8-bit), QLoRA (rank 8, 16 layers, lr 10^{-5}, batch 2, prompt masked, $\approx$4 epochs over the consolidation set). Domain: a family of small-item bin-packing distributions in which best fit is beatable by simple heuristics (headroom 0.3–0.5 points of excess; first fit ties or beats best fit), ten training variants and eight held-out variants that never enter any training set; six far families (Weibull, triangular, bimodal, OR-library) for far transfer; verification on 20 $\times$ 200-item instances; a *find* beats the better of best fit and first fit by 0.001 on the variant's training instances. Cycle 0 is a shared base generation (3 notebooks $\times$ 1,200 tokens per variant, an angle comment every 250 tokens). Cycles 1–5: *attract* trains a fresh adapter on the finds plus the top-40 candidates by training excess from its own lineage; *random* trains on a same-size random sample of valid candidates (the control that separates "trained on itself" from "trained on what is worth"); *base* generates without an adapter at the same seeds. On the cycle-5 lineage we then trained: *repel*

arm	cycle	candidates	mean excess	rel. to min(BF,FF)	best	distinct	copies
base	0	65	0.0633	+0.0322	0.0408	0.99	–
base	1	53	0.0562	+0.0251	0.0327	1.00	–
attract	1	235	0.0559	+0.0248	0.0319	0.98	10%
random	1	276	0.0708	+0.0397	0.0332	1.00	5%
base	2	42	0.0583	+0.0272	0.0317	0.97	–
attract	2	198	**0.0430**	+0.0119	0.0312	0.99	6%
random	2	274	0.0697	+0.0386	0.0310	0.99	8%
base	3	55	0.0644	+0.0333	0.0414	1.00	–
attract	3	214	**0.0516**	+0.0204	0.0311	0.99	7%
random	3	334	0.0653	+0.0342	0.0312	0.99	1%
base	4	57	0.0532	+0.0221	0.0323	1.00	–
attract	4	195	**0.0420**	+0.0109	0.0311	0.98	16%
random	4	276	0.0510	+0.0199	0.0307	1.00	0%
base	5	47	0.0556	+0.0244	0.0325	1.00	–
attract	5	226	**0.0386**	+0.0074	0.0311	1.00	1%
random	5	287	0.0695	+0.0383	0.0311	0.99	0%

Table 2: Battery C on the eight held-out variants (never in any training set; test instances): candidates per arm and cycle, mean test excess of valid candidates (cell means over variants), the same relative to the better of best fit and first fit, the best candidate, the distinct/valid ratio and the share of literal copies of the arm's training set. The value-filtered adapter lowers the mean without moving the best.

(DPO, $\beta = 0.1$, 80 steps, chosen = the attract set, rejected = the worst valid candidate of the same variant), *repel-anch* (the same plus a token-mean supervised term on the chosen side, $\alpha = 1$), *repel-early* (plain DPO, 20 steps), *qd* (supervised, one elite per behavioural niche of the verifier's per-instance profile, $k = 3$ niches per variant plus one seat for the best clone of the classic attractor) and *repel-mode* (anchored DPO whose rejected examples are the best-fit clones themselves). Hypotheses, on the held-out variants: C1 (primary), attract < base on the mean test excess of valid candidates (the prior moved on variants it never saw); C2, attract < random; C3, higher find rate; C4 (two-sided), the diversity cost; R1' repulsion adds to attraction; R2' the best moves; T1 the tail rate (share of candidates better than the classics) on far families; T2 functional collapse.

Attraction. Table 2 gives the held-out reading by arm and cycle (Figure 1). At the pre-registered final cycle (3), C1 was -0.0129 [-0.0317, +0.0027], $p = 0.109$, not supported at $\alpha = 0.05$, with the direction consistent (7 of 8 variants better) and the cycle-2 contrast supported (-0.0153, $p = 0.016$); a pre-registered extension to five cycles closed it: -0.0170 [-0.0268,-0.0079], $p = 0.008$, the attract arm ending at 3.86% mean excess, 0.74 points above the better classic heuristic, against 5.6–6.4% for the base. C2: attract vs random, -0.0138 ($p = 0.035$) at cycle 3 and -0.0309 ($p = 0.004$) at cycle 5: the value filter matters. The random control hurts at cycle 1 (+1.5 points over base: consolidating mediocre functions makes the next ones worse) and is neutral to harmful later. C3: strict finds are zero in every arm and cycle, and the best candidate per variant is nearly the same in every arm (≈3.1% excess, the level of best fit): the ceiling did not move. C4: no collapse over five cycles; the distinct/valid ratio is 0.97–1.00 throughout, literal copies of the training set 1–16% (attract); every adapter makes the model write 3.6–4× more functions per notebook. Far transfer: on six distributions of other families the final attract adapter equals the base on the mean (+0.001, $p = 0.69$; $n = 5$ of the six far families; the base arm produced no valid candidate in one) and beats the random adapter by -0.020 ($p = 0.016$); the random adapter is worse than the base there too (+0.017). What moved is the prior for the family it was trained on. The adapter of the last cycle was taught mostly tight-bin functions (24 of 40 of the form `1/(r**2+eps)`, 13 best-fit-like): consolidation

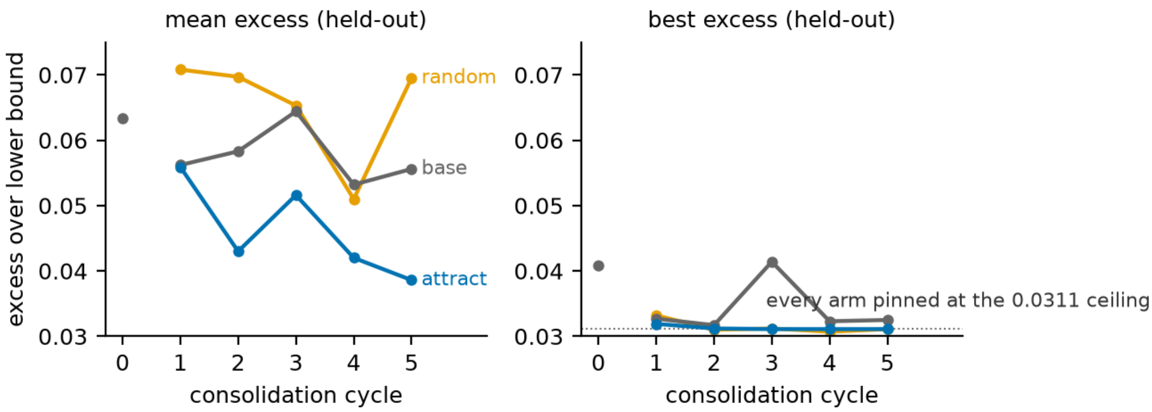

Figure 1: The title of this paper, as data (held-out variants, never in any training set; cycle 0 is the shared pre-consolidation base). Left: mean excess over the lower bound falls under attraction and nowhere else. Right: the best candidate never moves; every arm is pinned at the 0.0311 ceiling from the first cycle. Consolidation moves the mean, not the best.

	held-out variants (8)				far families	
adapter (cycle-5 lineage)	candidates	mean excess	at classic level	score levels	mean excess	better
base (none)	47	0.0556	20%	0.81	0.0838	
random	287	0.0695	15%	–	0.1004	
repel-early (DPO, 20 steps)	71	0.0721	30%	–	0.0994	
attract (SFT on value)	226	0.0386	72%	0.14	0.0803	
repel-anch (DPO + anchor)	144	**0.0311**	100%	0.07	**0.0758**	
qd (SFT on niche elites)	293	0.0598	25%	0.33	0.1012	
repel-mode (DPO from the attractor)	62	0.0792	2%	0.50	0.0935	
repel (DPO, unanchored)	13	0.1056	0%	–	–	

Table 3: The ladder of adapters. Held-out: mean test excess of valid candidates, share scoring exactly at the level of the better classic heuristic, distinct score levels per valid candidate; far families: mean excess and share of candidates better than the classics. The mass walks onto the known attractor; the tails live in the untrained base.

pulls the mass toward the best attractor the verifier has approved.

Repulsion. Unanchored DPO (loss 0.69 $\rightarrow$ 0.0015) degenerated: 1.1 closed functions per notebook (attract 9, base 2), 60% comment lines and recursive non-code; the few valid candidates were far worse than the base (10.6% vs 5.6% excess). Pushing the policy away from bad functions, unanchored, pushes it away from writing functions. Anchored (the same loss plus a token-mean supervised term on the chosen side; pre-registered), the repulsion writes again (144 valid candidates, +12 per cell over the base, $p = 0.02$) and adds to attraction: mean held-out excess 3.11% against 3.86% (-0.0074 [-0.0104, -0.0047], $p = 0.008$), which is exactly the level of the better classic heuristic. The best candidate does not move (0.0311 in both arms), and the reason is the finding: **every** held-out candidate of the anchored-repulsion adapter scores exactly at the best-fit level (all are monotone "fullest bin" variants), against 72% for attraction, 15–20% for the base and the random control, 30% for an early-stopped plain DPO (otherwise worse than attraction, +0.034, $p = 0.008$) and 0% for the degenerate one (Table 3). Hash-level diversity survives (1.00); functional diversity does not. On the far families the anchored adapter is the best arm on the mean (7.6% vs 8.4% base) because "be best fit" is good everywhere, and the only arm with no candidate better than the classics.

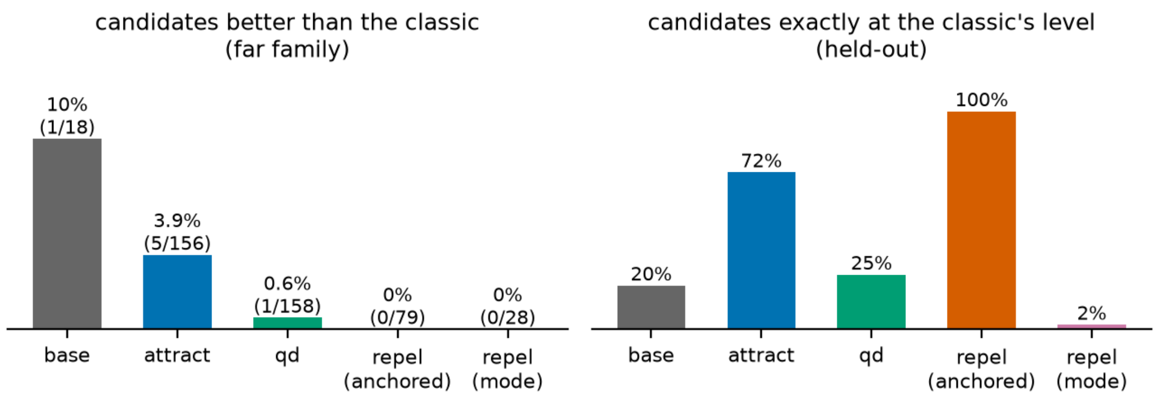

Figure 2: What consolidation costs. Left: the share of candidates *better* than the stronger classic heuristic on the far families, the tail where discovery lives, shrinks from 10% (base) to 3.9% (attraction) to at most 0.6% under every tail-preserving objective we built. Right: on held-out variants the consolidated arms concentrate *exactly at* the classic's level (attraction 72%, anchored repulsion 100%, a functional collapse), against 20% for the base. Absolute counts above the bars; the rates rest on 0–5 events per arm and are preliminary. Per notebook, attraction's absolute tail yield is higher than the base's (0.28 vs 0.08). Competence per candidate is bought with idiosyncrasy; total yield depends on production.

Keeping the tails. Two objectives built to keep the tails (pre-registered) recovered functional diversity and nothing else (Figure 2). The qd adapter has 25% of candidates at the classic level (attract 72%, $p = 0.004$) and 2.4× the distinct score levels, at a cost of +2.1 points of mean excess ($p = 0.008$), and its better-than-classic rate on the far families is *lower* than attraction's (0.6% vs 3.9%; T1, primary, not supported). Repulsion from the attractor itself removes the collapse entirely (2% at the classic level) and with it most of the value (7.9% mean excess, best 4.8%, worse than the base; no far tail). The diversity these objectives keep is diversity of the mediocre. Across every consolidation we ran, the per-valid-candidate rate of better-than-classic candidates on new families is highest in the untrained base (10% as a mean over families; pooled, 1 of 18 candidates) and falls with training (attract 3.9%, pooled 5 of 156; every other adapter $\leq$ 2%, pooled 0–1 events). The absolute yield points the other way: because consolidation multiplies valid production, attraction delivered five better-than-classic candidates against the base's one, 0.28 against 0.08 per notebook, across two families against one. Both readings rest on zero to five events per arm and are preliminary; what they jointly say is that consolidation concentrates each candidate toward the known good while producing many more candidates, so the per-candidate tail rate falls even where the total tail yield does not. The anchored objectives, by contrast, produced no better-than-classic candidate at all.

7 Discussion

What can be bought. Across the two studies the ledger is now complete for the interventions we could build. Variation is cheap: the interruption raises fresh surprise by more than a point and multiplies valid candidates three- to fourfold. Integration can be bought with the right memory: a schematic recap carried across resets is the only inference-time intervention that raised the document-level reading, and it did so reliably. Mass can be bought with the right training: a few cycles of consolidation on what the verifier approved move the candidate distribution of a never-seen variant onto the known good, and an anchored repulsion from the bad moves it there exactly. Progression and frontier cannot, by anything we tried: three injections, selection over the candidates, a verifier in the stream, a judge-gated review, five cycles of attraction, four kinds of repulsion and a quality-diversity consolidation all leave development near 2/10 and the verified ceiling where it was.

The bit distance (a conceptual model consistent with, not demonstrated by, these data). A generative model is a distribution, not a store; the question is not whether an answer is "inside" it but how much probability mass sits near it, how many bits of search separate the prior from the target. Inference-time stimulation reorganizes the surface of a fixed prior: it buys variation and, with a schematic memory, integration, but it does not move the mass. Weight consolidation moves the mass, and battery C shows its present reach: toward the best thing the verifier had already approved, within the family it was trained on, and at the cost of the tails where the rare better-than-classic candidates appear; objectives built to keep the tails kept diversity of the mediocre instead. This is also why the known successes of verified search [6, 9] pair an untrained proposal distribution with a hard evaluator and many samples rather than fine-tune the proposer. For discovery our results argue for a portfolio, the broad untrained prior to propose and the consolidated one to exploit, and for spending on selection.

Design rules. For long open-ended generation: damp repetition; interrupt with something new each time or reset the context; carry a compressed recap rather than the text; do not expect the whole to advance from any of this. For search with a verifier: spend on sampling and selection, not on stimulation; keep the verifier's verdicts out of the prompt or measure the fabrication they induce; if reliability is the goal, consolidate on what the verifier approved (and never without the filter); if discovery is the goal, keep the untrained prior in the loop.

8 Limitations

Ten premises and ten variants per battery; 8–30B base models at 8 bits; the document judge is a single LLM family at $k = 3$ and its absolute levels are low for every arm, so "integration 2.7" is a relative statement; the bin-packing distributions of batteries S and V have little headroom, which bounds what selection and feedback could show there (battery C's family was chosen for headroom for that reason, and its headroom is still small: 0.3–0.5 points); the oracle that writes recaps and questions is the generator itself and sometimes repeats itself, which the anti-copy ladder mitigates but does not remove; battery C is ONE training lineage of five cycles of forty examples on an 8B model with a rank-8 adapter; the eight held-out variants replicate the adapter, not the training procedure, and independent lineages (new generation seeds, new example sampling, fresh never-consulted held-outs, a stopping rule fixed in advance) are the required next step; its primary contrast missed α at the registered third cycle ($p = 0.11$) and closed only in a sequential-analysis sense at the extension to five ($p = 0.008$), the held-out set having been consulted at both looks; its base arm varies by seed; candidate counts differ by arm (47–334), so the observed best is not an equal-budget comparison of tails; the matched-k analyses (released appendix) find: at k=10 valid candidates per variant, attraction's best-of-k beats random's (0.0311 vs 0.0381 mean excess; both 8/8 variants) while the base reaches ten valid candidates in only one of eight variants (0.0449 there), so part of the base's small-sample best advantage is a production deficit; pass@k against the classics is zero for every arm at matched k; the far families are six (paired $n = 5$) and the tail rates rest on tens of candidates; the QD arm is a small implementation (three niches per variant), so "tail-preserving objectives keep the mediocre" is a statement about this implementation, not the class [14]; and the after the first cycle attract and random are two whole adaptive procedures rather than two selection criteria applied to one pool (a shared-pool one-shot control, top-40 vs random-40 from a single base pool, is registered and running with the new lineages); excluding the far family in which the base produced no valid candidate treats a production failure as missing data, so the paired far contrast is conditioned on validity; the selection criterion admits lucky candidates (training finds lose on test), which is why the primary measure is the mean and not the best. The negative results are resolution-bounded nulls, not proofs of absence; the tail finding in particular deserves replication at a larger sample and on a domain with more headroom.

9 Conclusion

Given memory, tension, selection and a verifier, the interrupted loop composes better and advances no further. A schematic recap raised judged integration relative to reset, while the pre-registered composite contrast was not supported; a standing question helps a little; selection and in-stream verification showed no detectable benefit on the near-saturated variants tested, and the latter teaches the model to imitate verdicts. Consolidating training-verifier-selected candidates into the weights is the one intervention that moved what the model writes on problems it had not seen, and it moved it toward what it already knew to be good; the combined anchored objective (supervised attraction plus repulsion from the worst) moved it all the way there and no further, though which half does the moving awaits a registered SFT-only control; keeping the tails kept the mediocre. In this lineage the untrained base had the highest per-candidate better-than-classic rate on the evaluated far families, while attraction's larger production delivered more absolute tail events; the observed best improved nowhere. Mean candidate quality can be bought; the observed best, so far, has not moved, and whether the procedure replicates is the question the registered independent lineages now running exist to answer.

A Reproducibility and experimental provenance

battery	registered	data already seen	primary	status
M/M+ (memory, tension)	2026-08-19	paper-1 batteries	schema vs verbatim	supported
S, V (selection, verifier)	2026-08-19	M	held-out gain	null; fabricated verdicts
C cycles 1–3	2026-08-21	none (new domain)	attract vs base (c3)	not supported ($p = 0.11$)
C cycles 4–5	2026-08-22	c3 result	attract vs base (c5)	sequential, $p = 0.008$
C-repel, N (tails)	2026-08-22	C	keep the tails	degenerate / not kept
Matched-k analyses	2026-08-24	all C	best-of-k, pass@k	reported
C-rep (3 lineages)	2026-08-26	all above	procedure replicates?	running

Table 4: Order of registrations and looks. The cycle-5 result is a prospective sequential extension, not an independent confirmation; the independent lineages with never-consulted held-outs are the confirmation battery.

Code, run data, judgments, pre-registrations and the dated laboratory notebook are in the repository of the companion study (`https://github.com/RobertoOno/interrupting-the-loop`): the oracle arms (`schema_reseed`, `anomaly_reseed`, `agenda_reseed` in `scripts/dream_run.py`), battery S (`scripts/evolve_interrupt.py`), battery V (`scripts/problem_loop.py`), battery C and its extensions (`scripts/consolidate.py`, `dpo_lora.py`, `run_c.sh`, `run_c_ext.sh`, `run_c_repel*.sh`, `run_n.sh`) and the analyses that generate the tables (`docs/APPENDIX_GEN.md`, `APPENDIX_S.md`, `APPENDIX_V.md`, `APPENDIX_C*.md`, `APPENDIX_N.md`). Judging used `anthropic.claude-opus-5` on Amazon Bedrock (20–21 August 2026) under the prompts of the companion study; fine-tuning used `mlx_lm` (QLoRA on the 8-bit model, peak 14 GB, about 8 minutes per adapter) and a small DPO trainer in MLX released with the code. Every hypothesis, amendment and result is dated in `docs/PLANO.md`.

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